Problem 1

Calculate the time derivative of the following rotation matrices. The angular velocity $\boldsymbol{\omega}$ is expressed in the fixed frame.

(a) $\boldsymbol{\omega} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ rad/s, $\boldsymbol{R} = \begin{bmatrix} 0.5 & -0.866 & 0 \\ 0.866 & 0.5 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.

(b) $\boldsymbol{\omega} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$ rad/s, $\boldsymbol{R} = \begin{bmatrix} 0.5 & -0.866 & 0 \\ 0.866 & 0.5 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.

(c) $\boldsymbol{\omega} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ rad/s, $\boldsymbol{R} = \begin{bmatrix} 0.5 & -0.866 & 0 \\ 0.866 & 0.5 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.

(d) $\boldsymbol{\omega} = \begin{bmatrix} 0.91340 \\ 0.63240 \\ 0.09754 \end{bmatrix}$ rad/s, $\boldsymbol{R} = \begin{bmatrix} -0.62170 & -0.02073 & -0.78300 \\ -0.13850 & -0.98100 & 0.13590 \\ -0.77090 & 0.19300 & 0.60700 \end{bmatrix}$.

[Work not yet added: paste solution here]

Problem 2

A quadcopter is flying around with an inertial measurement unit (IMU) attached 10 cm offset from the center of mass (Figure **??**). An external observer measures the position, orientation, linear and angular velocity, and linear acceleration of the quadcopter relative to a fixed ground reference system (origin 0). What is the linear velocity and acceleration, and angular velocity measured by the IMU in the body frame of the quadcopter at this instant? The angular velocity remains constant. *Hint: Do not forget gravity.*

$$ {}^0\boldsymbol{d}_{01} = \begin{bmatrix} 20 \\ 15 \\ 10 \end{bmatrix} \text{m}, \quad {}^0\boldsymbol{R}_1 = \begin{bmatrix} 0.5 & -0.866 & 0 \\ 0.866 & 0.5 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad {}^0\boldsymbol{v}_{01} = \begin{bmatrix} 10 \\ 5 \\ 0.25 \end{bmatrix} \text{m/s}, $$

$$ {}^0\boldsymbol{a}_{01} = \begin{bmatrix} -1 \\ 0 \\ 0.1 \end{bmatrix} \text{m/s}^2, \quad {}^0\boldsymbol{\omega}_{01} = \begin{bmatrix} 0 \\ 0 \\ 2.3 \end{bmatrix} \text{rad/s}. $$

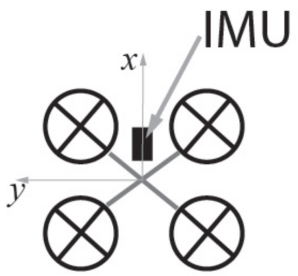


Figure 1: IMU offset from the center of mass.

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Problem 3

Consider a three-link manipulator with joint angles $\boldsymbol{\theta} = [\pi/3 \;\; \pi/6 \;\; \pi/6]^\intercal$ rad, joint angle rates $\dot{\boldsymbol{\theta}} = [0.5 \;\; -1.0 \;\; 1.5]^\intercal$ rad/s, a stationary base, and the following DH table. Assume that the angle rates remain constant. What are the angular velocities $\boldsymbol{\omega}_i$, linear velocities $\dot{\boldsymbol{d}}_{0,i}$, and accelerations $\ddot{\boldsymbol{d}}_{0,i}$ at each frame (i.e., $i = 1, 2, 3$)? In addition, construct the velocity Jacobian. *Hint: The velocity of the end frame is* $\boldsymbol{J}_{\text{v}}\dot{\boldsymbol{\theta}} = [-0.1924 \;\; 0.2498 \;\; -0.3580 \;\; 0.4330 \;\; -0.2500 \;\; 0.5000]^\intercal$.

Link	a	d	α	θ
1	0	1	$\pi/2$	$\star$
2	0.5	0	0	$\star$
3	0.3	0	0	$\star$

[Work not yet added: paste solution here]

Appendix A: Code